

01 Research Interests

Reliable and efficient video understanding.

I am a Ph.D. candidate in Brain & Cognitive Engineering at Korea University and a student researcher at KIST AI-Robotics. I study reliable and efficient video understanding: identifying anomalous behavior, understanding its temporal context, and explaining it with visual evidence.

My work connects video anomaly detection and understanding, adaptive video sampling, action recognition, and vision-language models (VLMs). Through public-safety research at KIST, I work on contextual anomaly detection and behavior analysis for real-world surveillance video.

Recent work includes BUSTER for VLM-guided anomaly detection with adaptive sampling, Where and What for contextual anomaly detection, and Look Closer for action-guided proficiency estimation.

Current focus: why vision-language models hallucinate in video anomaly understanding, and how selective visual verification can make their explanations more faithful.

- Video understanding
- Video anomaly detection & understanding
- Adaptive video sampling
- Efficient vision-language models
- VLM hallucination & reliability
- Temporal & contextual reasoning
- Action recognition & human motion analysis
- Video surveillance & public safety

02 Education

Mar 2023 – Present Seoul, KR	Ph.D. Candidate, Brain & Cognitive Engineering Korea University - Expected graduation: March 2027. - Study contextual video understanding inspired by human cognitive processes.
Aug 2015 – Feb 2020 Seoul, KR	M.Eng., Electronic Engineering Kwangwoon University - Thesis: A Study on Passive Target Tracking in Three-Dimensional Indoor Environment by Using Deep Neural Network. - Completed 19 months of mandatory military service.
Mar 2013 – Aug 2015 Seoul, KR	B.Eng., Electronic Engineering TRANSFER Kwangwoon University Transferred in March 2013 from the National Institute for Lifelong Education after completing B.Eng. coursework in Computer Engineering from March 2011 to February 2013.

03 Research & Work Experience

Jan 2023 – Present Seoul, KR	Student Researcher CURRENT KIST AI-Robotics Institute · Vision Intelligence Group Video anomaly understanding · Surveillance video · Vision language models Develop public safety video systems spanning action recognition, contextual anomaly detection, and vision language models that explain anomalous behavior. [1] [2] [3] [4] [5]
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Jan 2022 – Feb 2023
Seoul, KR

Research Intern

[KIST AI-Robotics Institute](#) · [Vision Intelligence Group](#)

Began surveillance video research through urban public safety projects and national research programs.

Oct 2021 – Dec 2021
Sejong, KR

Researcher

[The Korea Transport Institute](#)

Contributed to autonomous bus development through SLAM, control center systems, policy research, and coordination.

04 Research Grants & Projects

~\$24M
Five year program

Real-Time Intelligent Surveillance for Public Safety

NATIONAL R&D

[MSIT & MOTIE](#) · [with Yonsei University and the Korean National Police Agency](#)

Video anomaly detection that uses context across city scale CCTV control centers with about 1,000 cameras, validated in Anyang City.

~\$9.5M
Five year program

Mobile Platform for Active Safety Services

[Korean National Police Agency & MOTIE](#)

Government research program using mobile platforms for proactive public safety services.

New
initiative

Anomaly Detection in Correctional Facilities

[KIST AI-Robotics](#)

Detecting inmate anomalous behavior in custodial environments.

05 Selected Publications

- [1]** Ahn D., Kim H., Han J.-H., Ham B., Choi H., Kim I.-J., Kim H. **ACCEPTED**
Look Closer: Action-Guided Dense Visual Dynamics for Proficiency Estimation.
[CVPRW 2026](#) · [Multimodal Human Motion Analysis](#) · Pitch and Poster Presentation
- [2]** Ahn D., Choi Y., Han J.-H., Kim I.-J., Kim H. **ACCEPTED**
BUSTER: Adaptive Sampling for VLM-Guided Unsupervised Video Anomaly Detection.
[British Machine Vision Conference \(BMVC\), 2026](#)
- [3]** Ahn D., Jo Y., Lee U.-S., Park H., Han J.-H., Kim H. **SUBMITTED**
Learning Robust Representations for Few-Shot Action Recognition with Frame-Level Ambiguities.
[IEEE Access, 2026](#)
- [4]** Ahn D., Jo Y., Kim D., Nam G. P., Han J.-H., Kim H. **PUBLISHED**
Where and What: Contextual Dynamics-Aware Anomaly Detection in Surveillance Videos.
[IEEE Transactions on Image Processing, 2025](#) · IF 13.7 · Top 2.2% in E.E. Eng.
- [5]** Kim D., Ahn D., Jo Y., Park H., Lee S., Kim H. **PUBLISHED**
A Foundational Research Framework for Real-World Abandoned Object Detection: Train-Free Baseline and a Standardized Benchmark.
[Expert Systems with Applications, 2025](#) · 130658
- [6]** Zhou B., Ahn D., Lee J., Sun C., Ahmed S., Kim Y. **PUBLISHED**
A Passive Tracking System Based on Geometric Constraints in Adaptive Wireless Sensor Networks.
[Sensors, 18\(10\), 3276, 2018](#)
- [7]** Ahn D., Kim N. M., Park J. H., Kim Y. O. **PUBLISHED**
2D Indoor Map Building Scheme Using Ultrasonic Module.
[J. Korean Inst. Comm. and Info. Sci., 41\(8\), 986–994, 2016](#)

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Awards & Honors

Dec 2025	AIR Outstanding Research Award KIST AI-Robotics Institute In recognition of outstanding research achievements during 2025.
Feb 2025	Excellent Paper Award, Graduate School Korea University For outstanding research performance and publication of an excellent paper during doctoral studies.
Nov 2024	Outstanding Oral Presentation Award KIST Academia and Research Convergence Conference For the presentation "Multi-Aspect Contextual Dynamics for Anomaly Detection in Surveillance Videos."
Jul 2021	Division Commander's Commendation Capital Mechanized Infantry Division For a combat development proposal on reconnaissance drone AI improvement and acoustic detection using AI and big data.

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Teaching

Sep – Dec 2018	Senior Capstone Project Kwangwoon University · Instructor: Prof. Youngok Kim Assisted in managing and supporting the Capstone Design Project as a TA for 50+ students.
Mar – Jun 2017	Computer Architecture Kwangwoon University · Instructor: Prof. Youngok Kim Graded assignments and provided feedback for 30+ students with Q&A sessions.

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Technical Skills

Programming	Python	C	Java	Git	Linux	
Machine Learning	PyTorch	TensorFlow	Keras	NumPy	Pandas	Matplotlib
Sensors	Ultrasonic	Beacon	RF	Wi-Fi	3-axis Accelerometer	
Embedded & IoT	Raspberry Pi	Arduino				